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CUSTOMER SHOPPING BEHAVIOR

SQL Portfolio — Questions & Answers

10 SQL queries · Aggregation, Filtering, Subqueries, CASE, CTEs & Window Functions

Q1. What is the total revenue generated by male vs female customers?

```sql
SELECT gender, SUM(purchase_amount) AS revenue
FROM customer
GROUP BY gender;
```

Q2. Which customers used a discount coupon but still spent more than the average purchase amount?

```sql
SELECT customer_id, purchase_amount
FROM customer
WHERE discount_applied = 'Yes'
  AND purchase_amount > (SELECT AVG(purchase_amount) FROM customer);
```

Q3. Which are the top 5 products with the highest average review rating?

```sql
SELECT item_purchased, ROUND(AVG(review_rating::numeric), 2) AS average_review
FROM customer
GROUP BY item_purchased
ORDER BY average_review DESC
LIMIT 5;
```

Q4. Compare the average purchase amount between Standard and Express shipping

```sql
SELECT shipping_type, ROUND(AVG(purchase_amount), 2)
FROM customer
WHERE shipping_type IN ('Standard', 'Express')
GROUP BY shipping_type;
```

Q5. Do subscribed customers spend more? Compare average spend and total revenue between subscribers and non-subscribers

```sql
SELECT subscription_status,
       COUNT(customer_id),
       AVG(purchase_amount) AS average_spend,
       SUM(purchase_amount) AS total_revenue
FROM customer
GROUP BY subscription_status
ORDER BY average_spend, total_revenue DESC;
```

Q6. Which 5 products have the highest percentage of purchases with discounts applied?

```sql
SELECT item_purchased,
       ROUND(100.0 * SUM(CASE WHEN discount_applied = 'Yes' THEN 1 ELSE 0 END)
             / COUNT(*), 2) AS discount_rate
FROM customer
GROUP BY item_purchased
ORDER BY discount_rate DESC
LIMIT 5;
```

Q7. Segment customers into New, Returning, and Loyal based on their total number of previous purchases, and show the count of each segment

```sql
WITH customer_type AS (
  SELECT customer_id, previous_purchases,
    CASE
      WHEN previous_purchases = 1 THEN 'New'
      WHEN previous_purchases BETWEEN 2 AND 10 THEN 'Returning'
      ELSE 'Loyal'
    END AS customer_segments
  FROM customer
)
SELECT customer_segments, COUNT(*)
FROM customer_type
GROUP BY customer_segments;
```

Q8. What are the top 3 most purchased products within each category?

```sql
WITH item_counts AS (
  SELECT category, item_purchased,
         COUNT(customer_id) AS total_orders,
         ROW_NUMBER() OVER (PARTITION BY category
                            ORDER BY COUNT(customer_id) DESC) AS item_rank
  FROM customer
  GROUP BY category, item_purchased
)
SELECT item_rank, category, item_purchased, total_orders
FROM item_counts
WHERE item_rank <= 3;
```

Q9. Are customers who are repeat buyers (more than 5 previous purchases) also likely to subscribe?

```sql
SELECT subscription_status, COUNT(customer_id) AS repeat_buyer
FROM customer
WHERE previous_purchases > 5
GROUP BY subscription_status;
```

Q10. What is the revenue contribution of each age group?

```sql
SELECT age_group, SUM(purchase_amount) AS total_revenue
FROM customer
GROUP BY age_group
ORDER BY total_revenue DESC;
```